# Data Transformation

## Assignment

PW SKILLS

**Objective**

Prepare the given dataset for Machine Learning by applying:

- Encoding
- Normalization
- Standardization (Scaling)
- Preprocessing Pipeline

**Dataset:** Employee_productivity_Dataset

**Instructions**

- Use Pandas and Scikit-learn
- Perform steps in order
- Display output after each question
- Write short explanations where asked

**Assignment Questions**

**Q1. Handle Missing Values (Basic Cleaning)**

Fill missing values using the following:

- Age → Median
- Salary → Mean
- Hours_Worked_Per_Week → Median
- Performance_Score → Mean

Display the dataset after handling missing values.

**Q2. Label Encoding**

Convert these columns into numeric values using Label Encoding:

- Gender
- Department

Show the updated columns.

**Q3. One-Hot Encoding**

Apply One-Hot Encoding on:

- Work_Mode
- Location

Display the dataset and check how many new columns are created.

**Q4. Normalization (Min-Max Scaling)**

Normalize the following columns:

- Salary
- Hours_Worked_Per_Week

Use MinMaxScaler and display results.

**Q5. Standardization (Scaling)**

Apply StandardScaler on:

- Age

- Projects_Completed

Display the transformed values.

**Q6. Compare Scaling Methods**

Apply both:

- MinMaxScaler
- StandardScaler

On the Salary column.

Show both results side by side.

**Q7. Build Preprocessing Pipeline**

Create a pipeline that:

- Applies encoding to categorical columns
- Applies scaling to numerical columns

Use:

- ColumnTransformer
- Pipeline

**Q8. Apply Pipeline**

Apply the pipeline on the dataset.

Display:

- Transformed dataset
- Shape of final dataset

**Q9. Conceptual Question**

Why is scaling important in Python?

**Q10. Conceptual Question**

Why do we convert categorical data into numerical form?